

Andrew Ton

Hands-on Manager/Architect/Developer

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Clearance: Public Trust, US Citizen

Summary: Expert on **backend** (24+ years), expert **frontend** (12+years), expert **database** (30+ years), expert **AWS Cloud** (4+ years), experienced team lead/architect (8+ years). **Experience with LLM AI RAG (Regenerative Augmented Generation) Design and Development**. In summary, **5 expertise in 1 employee**, 33 years hands-on.

Experienced Hands-on Architect, Manager, and Lead Developer with over 33 years of dedicated software development expertise. Proven track record in leading high-performance teams and delivering cutting-edge solutions for the government and commercial sector. Specialized in full-stack development, backend architecture (Java/Spring-boot), UI design and implementation (**Angular** V8 to V16, TypeScript 5, **React** 15-18, NgRx, Redux, Mobx, Kendo-lib, Material-lib), Tailwind CSS, test with Karma/Jasmine, Bootstrap, NG-Bootstrap, and large-scale database design (Oracle/Mongo and all existing databases) and implementation. Experience with Cloud: AWS and **Azure**, both serverless and EC2/EKS cloud lift-shift. Experience with CSS/SASS and WCAG 2.0. Seeking leadership roles and hands on mentorship and implementation in projects that leverage modern technologies and foster innovation.

Video Demo

(youtube videos, safe to click):

- 1) Startup, Solo Developer (a to z): Frontend Developer, Backend Developer, AI developer, Requirement, dba/database architect, Hardware architect (ha/clustering/proxy/https), Network architect(router/switches/LB), Security Architect(Firewall/IntrusionDetection), RAID, deployment, founder with 70% ownership),

TutaDates:https://youtu.be/HcclBurPtew?is=IhfhV-Cy5Y1_m9JM

Video Demo of TutaNotes and TutaPhotos:

Part 1 of 2: <https://youtu.be/3ZSUKApMIUU>

Part 2 of 2: <https://youtu.be/KhcARpOEhrY>

Please visit LIVE/Production site at <https://tutamall.com>

(login: **demo**, password: (not required for demo) [Coming soon TutaNetwork (like LinkedIn) and TutaMarketPlace (like Ebay)]

[TutaMall.com](https://tutamall.com) can support 1.2 parallel users, Facebook 16 million (but need 9,000 engineers/Additional 5,000 engineers for instagram, where I built tutamall.com solo, a to z, fullstack, db, network, hardware, firewalls)

- 2) **AI RAG** with TutaNotes Demo: <https://youtu.be/6JdPIO1zMdk>

- 3) **Live coding demo:** and Deployment of **React** (with Kendo Grid) with **latest AWS CDK script for serverless:** Lambda+Gateway+stepFunction+S3+Dynamo demo:

https://youtu.be/C0lt_HSkJE4

- 4) Live coding demo: **AWS CDK script for CI/CD CodePipeline:**<https://youtu.be/5KM10xDvecQ>

- 5) Live coding demo: **AWS CDK script for SES/Email** <https://youtu.be/wDkVu57u0w8>
- 6) Live coding demo: My **Backend Spring boot** from scratch: <https://youtu.be/p6JnNzHb1Oc>

demo: **Advance UI: NgRx** (UI Redux): <https://youtu.be/HZatG3VOVKM> 7) My real Production Angular 16 UI Demo (this video demo my cutting-edge custom design, separate PODS/**Kubernetes** for angularJS and Angular16/Kendo, yet coexist seamless navigation using SSO/**JWT Tokens**. Similar Pentagon/Crystal city Angular 8 project, with a large 30+ team, unable to accomplish what I did here, solo design and solo implementation.): <https://youtu.be/Fbod7l4Xkkl>

Other Demo Video/Image/Doc of UI Frontend, Backend and Database experiences

1. Samples of **UI work** can be view at:

<https://drive.google.com/file/d/1n6u8wCw1zGYAY01eBPLDtsybwPYnvFHx/view?usp=sharing>

2. **Cover Letter**

<https://drive.google.com/file/d/1SOYAQkoMJp6VbKqavAH50aKyCnLUREyg/view?usp=sharing>

3. Summary of Important Tech Stacks I worked on extensively:

https://drive.google.com/file/d/1uk5onhezf0HP_1oQLw4zr_8By6Z9tRSj/view?usp=sharing

1991 to 2024, 33 years complete work history is at this link:

<https://drive.google.com/file/d/1naqG5ygyzxlZ-ur2lpfpN6kraK6v1hzJ/view?usp=sharing>

Professional Experience

TutaMall, Startup, Architect/Developer

April 2024 - Present

- Startup: Founder, Solo Developer (a to z): Frontend Developer, Backend developer, AI Developer, Requirement, DBA/database Architect, Hardware Architect (ha/clustering/proxy/https), Network Architect(router/switches/LB), Security
- Architect(Firewall/IntrusionDetection), RAID, deployment, founder with 70% ownership) **TutaDates:**https://youtu.be/HcclBurPtew?is=IhfhV-Cy5Y1_m9JM
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Team Lead/Architect, Bloomberg/FMIS Senate Financial Project

March 2022 - April 2024

- Architected and developed a pioneering hybrid modernization solution for Senate financial operations, integrating Angular 8-16/Kendo with existing AngularJS modules seamlessly. • Architected and Hands-on development solo of complete **AI LLM RAG - Retrieval Augment Generative** - Extracting CSV file Data Transform to Knowledge Graph, query with LangChain, Creating a Vector Index, Creating Embeddings and Populating into DB
- Demonstrated expertise in JWT Token SSO implementation, ensuring secure and unified user experiences across microservices deployed on OpenShift PODs.
- Led a team to successfully migrate over 45 AngularJS components to Angular 16, improving system performance and maintainability.
- Backend of our Financial Project is 100% java Spring Boot 1.0 (before 2017) existing, but I personally added new 8 new set of spring boot: controller (@RestController/ @Autowired), services (@Services), repositories (JpaRepository), entities (@Entity). This includes 3 new tables, many to many joins.
- Orchestrated backend solutions utilizing **Azure** and AWS platforms, incorporating AWS Lambda/S3/Dynamo/StepFunctions/SignedUrl/CloudFormation.
- Spearheaded frontend development with React and Redux-toolkit, AG-Grid, integrating state-of-the-art features for enhanced user interaction and data management. • Backend NestJS/TypeORM (similar to Spring Boot Data JPA)
- Tailwind CSS, test with Karma/Jasmine.
- Built DevOps CICD pipeline for AWS CDK.

Manager/Architect/Team Lead/Developer, Koniags/Expression Inc., Washington DC

April 2017 - March 2022

- Directed an 8-member team through the development lifecycle, from project initiation to deployment, achieving high praise from government clients for **React** UI sophistication and performance.
- Security API integrations (Authentication/Authorization, PKI/JWT/OAuth 2/OIDC, KeyCloak). Installed KeyCloak (start Aug 2021), Identity and Access Management (IAM), from scratch, onto my team project (I was team lead) with React Frontend and Javascript Backend. Fully tested KeyCloak by test team and deployed successfully to SCIF (I personally deploy at SCIF) and it's on production since Dec 2021.
- Designed and implemented complex Oracle database structures, optimizing data retrieval and management for military applications on the East Coast.
- One of backend of Jets is 100% java Spring Boot 2.5.0. I personally wrote, from scratch, entire Spring Boot project, again, from scratch, complete with 35 tables, new set of spring boot: controller, services, repositories, entities for 35 new tables, with many to many joins. Complete with transaction start/stop and rollback (@Transactional), in java.
- Add additional functionality to existing **Liferay DXP** backend for Expression. Integrated third-party systems using Liferay's Service Builder framework and REST APIs. Architected and developed custom portlets using Spring MVC and JSP for document management system serving 2,000+ internal users
- Engineered Angular 8/9 applications with Reactive Forms, RxJs, and Material UI, facilitating seamless migration from AngularJS.
- Backend of our Financial Project is 100% java Spring Boot 2.0. They exist, but I personally made many update, add new table and column and new components list here for new features/stories: new controller (@RestController/ @Autowired), new services (@Services), repositories(JpaRepository), new entities (@Entity).
- Implemented agile methodologies and peer code reviews to enhance team productivity and code quality.

Senior Software Engineer, TechFlow Employee, Sorrento Valley, CA

January 2010 - April 2017

- Designed and implemented medical portals using AngularJS, HTML5, and Spring MV ensuring compliance with industry standards and regulations.
- **Liferay** backend is built on Spring Framework, complete with Dependency Injection, ORM, and transaction, JPA 2.2/Hibernate 5.1(same as Spring Boot 1.0). So I implement Liferay java controller, entities, repositories, and service layer as in Spring Boot. Spring boot like annotations, ManyToMany entities, HQL, and Criteria.
- Successfully re-architected and migrated large-scale government projects from legacy Liferay systems to newer versions, improving system efficiency and usability.
- Proficient in JSF, JavaScript, PHP, Python, and Hibernate, contributing to various enterprise software projects with a focus on backend development and integration.
- Developed JSF and Portlet (JSR 168/286) applications for USCIS modernization projects,

- integrating with Tibco/Seibel via WebService WSDL/SOAP for complex screen functionalities. •
- Designed and developed portals and custom JSF components for IRS projects using IBM Portal and WebSphere Application Server, ensuring compatibility and compliance with federal standards.
- Back end is 100% Spring Boot, I did extensive coding in Hibernate and JPA. •

January 2010 - October 2010

- Led the development of the Automatic Threat Prioritization project, designing and •
- Implementing the database architecture from scratch using Oracle and Java. Back end is 100% Spring Boot, I did extensive coding in Hibernate and JPA.

[Dozen of jobs from 1991 to 2010, all software engineer jobs, not include to keep resume brief, but [click here for complete list of all jobs history](#)]

Certifications

- AWS Certified Practitioner (AWS CLF-C01) - **Expired: June 30, 2026**
https://www.credly.com/badges/a21ed8a483e741dd-819f-36272c6e9f49/public_url

Education

- BS in Computer Science, Virginia Polytechnic and State University, Blacksburg, Virginia •
- Graduated in May 1991, Bachelor in Computer Engineering (Major GPA 3.74 of 4.0) and Minor computer science (Minor GPA 3.82 of 4.0)
- Official Grade Transcript from VA Tech available upon request.

Technical Skills

- Languages: Java, Python, JavaScript, TypeScript, SQL
- Frameworks/Libraries: Spring Boot, Hibernate, Angular, React, Redux, Node.js •
- Databases: Oracle, MongoDB, MySQL
- Tools/Platforms: AWS, **Azure**, Docker, Kubernetes, OpenShift, Git, Jenkins